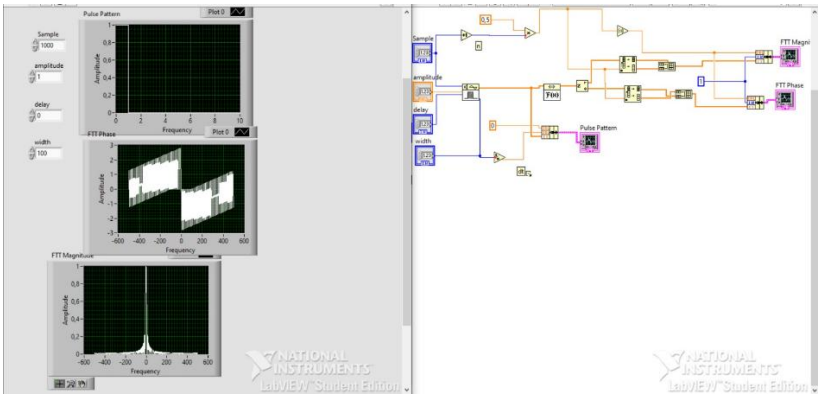
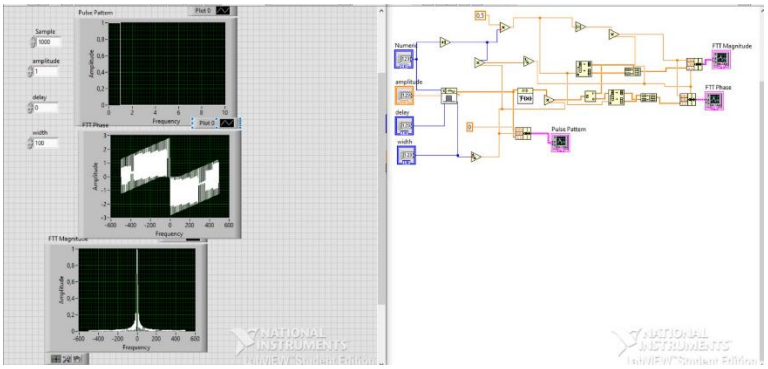
1) When 3.1.1 circuit was constructed and then simulated given the same pulse signal as the previously plotted one, we see that the FTT has only positive values whereas FT has both negative and the positive numbers. The FTT can be defined as the separated version of the FT that was done previously. The FFT takes the positive half of the samples and then takes the negative part of the sample.



2) Split 1D Array blocks index input is 1000 samples. These 1000 samples then get halved by the splitter into two 1D arrays with 500 samples.

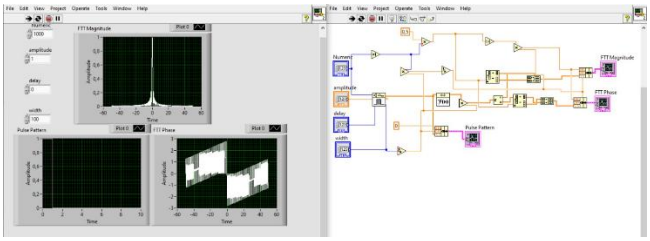
3) The value of the first Bundle Block input is 500.5. The graph cannot start from 0 so it is shifted by 0.5. That is why it is a double value.

4) The graphs are not same. The maximum value of magnitude is also scaled by 1/100 when we scaled the FFT results by 1/100.



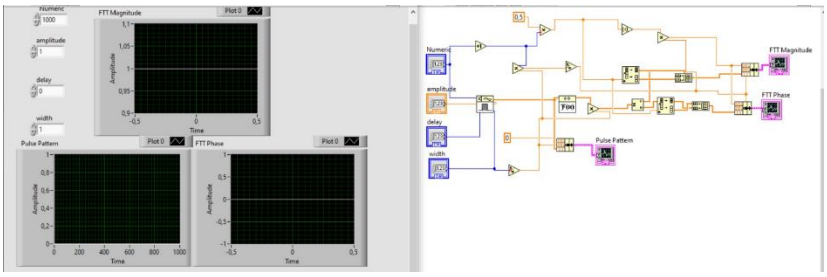
5) The x axis of “FFT Magnitude” and “FFT Phase” graphs are representing frequency. The values are directly proportional.

6) The range of the graphs differ as FFT's range is roughly between 0-1000, whereas the range for the FT is between -50 and 50. The frequency domain is scaled by 1/10.

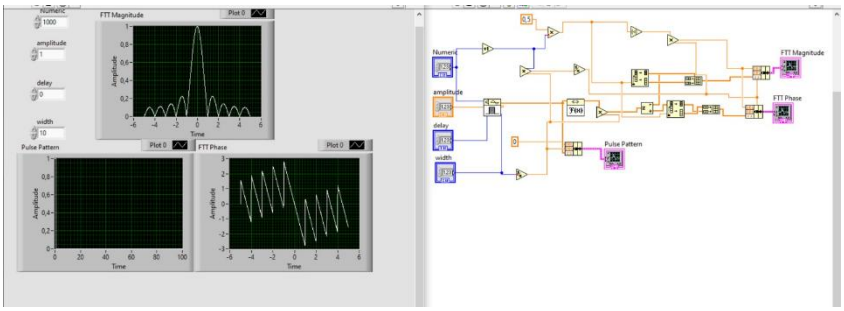


7) The width is brought down to 1 from 100, the plot became a line and when the width was 10 the plot that was done in pre-lab was reached. When the width increases the plot starts to behave like an impulse function. Update your VI by applying the frequency interval to graphs as shown in figure below and observe the results

Width = 1



Width = 10



Width = 1000

